Principles of Computer Programming (2440099) — Week 1 Lab

Lab 0 — Prove Your Toolchain Works

Time	One lab session
Weight	Pass/fail, but you cannot submit Project 1 until this is passed.

You will not write any interesting code today. The point is that from next week onwards, nobody in this room loses an hour to a broken compiler.

What to hand in

A single PDF or markdown file with four screenshots and two short answers.

1. A green build

Run, in the VS Code terminal:

```
cmake --preset debug
cmake --build --preset debug
```

Screenshot the last few lines showing it linked game with no errors.

2. All tests passing

```
ctest --preset debug
```

Screenshot the summary line showing all tests passed.

3. The debugger stopped at a breakpoint

Open src/main.cpp, find the line marked <<< BREAKPOINT HERE, and click in the gutter to the left of the line number to place a red dot. Press F5. Press an arrow key in the terminal. Execution stops.

Screenshot the whole VS Code window showing: the yellow highlighted line, the Variables panel, and the Call Stack panel.

> **ANSWER IN ONE SENTENCE** What value does key hold, and why is it that number and not the number of the letter you pressed?

4. A sanitizer report you caused on purpose

In src/main.cpp, find the DELIBERATE BUG block and uncomment the first line:

```
room.data()[room.cell_count()] = 'X';   // writing one past the end
```

Rebuild and run. The program will stop with a long red report.

Screenshot the first fifteen lines of it.

> **ANSWER IN TWO OR THREE SENTENCES** The report names an error type, a source file, and a line number. Which line does it blame, and how far past the end of the array did the write go? Then put the comment back.

Check you are ready

Before you leave the lab, run this and make sure it passes — it is exactly what the automatic grader will run on your submissions:

```
cmake --preset strict && cmake --build --preset strict && ctest --preset strict
```

This build treats every warning as an error. If it fails now, it will fail on every project you submit.

> **BEFORE YOU LEAVE** Confirm all four screenshots and both written answers are saved together in one file, and that you re-commented the DELIBERATE BUG line before your final green-build screenshot (or take that screenshot before uncommenting it).

One more check — not graded

Confirm Cline is installed (docs/SETUP_WSL2.md step 7) and that you can see a DeepSeek key in its settings panel.

> **DO NOT USE IT YET** Do not send Cline a prompt today — AI assistance is off for weeks 1–3 by course policy. Installing it now just means it's ready when that changes in week 4.